

Neeraj Jadhav

Education

- 2024-26 **Master of Science**, *Computer Science*, Specialization: Computing Systems
Master's Thesis: Can LLMs do program behavior analysis and optimization?
Prof. Santosh Pande, Georgia Institute of Technology, GPA: 4.0
- 2020-24 **Bachelor of Technology with Honors**, *Computer Science and Engineering*
Minor in Artificial Intelligence and Data Science
Indian Institute of Technology, Bombay, (Major) CGPA: 9.34, (Minor) CGPA: 9
Key courses: Algorithms, Operating Systems, Compilers, AI and ML, RL, Optimization Models

Publications

- **Phaedrus: Exploring Dynamic Application Behavior with Lightweight Generative Models and Large-Language Models** (Submitted to PLDI, 2025)
Bodhisatwa Chatterjee, Neeraj Jadhav, Sharjeel Khan, and Santosh Pande

Honors & Awards

- Recipient of **Masters Graduate Fellowship in CS** by *Georgia Institute of Technology* ['24]
- Amongst **top 10** teams in **Algorithmic Trading** contest organized by Web and Coding Club ['22]
- Secured **46th/150k** and **186th/870k** national ranks in **JEE Advanced** and **Main** exams resp. ['20]

Experience

- Summer '23 **Quantitative Research Internship**, *Quadeye Securities LLP*, Gurugram, India
- Implemented a **trading simulator** and optimization scripts in **R** to emulate **high-frequency** trading
 - **Optimized** the strategy using **black-box methods** improving **time spent by 80%** and **PnL by 2%**
- Winter '22 **Software Engineering Internship**, *FinIQ Consulting*, Pune, India
- **Enhanced** the proprietary financial coding language **FinSPL**, by adding **operational parallelism** and **efficient** data structures for information **storage** and **retrieval** of streaming **market indices**
- Summer '22 **Automated Orchestration using LLMs**, *IBM Research*, Gurgaon, India
- **Streamlined** business task execution using **LLMs** and **process mining** to optimize **workflow efficiency**
 - Pipelined **data extraction** from **conversation logs** by fine-tuning **GPT2** for seamless **API calls**

Research

- Fall '23 **Robust Celltree : A Paradigm for Distributed Data Repositories**, *Bachelor's Thesis*
Prof. Manoj Prabhakaran (IIT Bombay), *Prof. Indranil Gupta (UIUC)*
- Designed a **new data structure** with **safety** guarantees to **minimize** computational **load and time**
 - Enhanced **blockchain** with **proof-of-stake** and modified **consensus**, achieving high **latency gains**
- Fall '23 **mpC : A Secure Multiparty Computation Language**
Prof. Manoj Prabhakaran (IIT Bombay)
- Developed a **language**, with backend in **C++**, for **abstracting out cryptographic protocols**
 - Secured **communication with TLS & reduced** compilation **time** for **rapid testing** of new protocols

Projects

- Fall '24 **Benchmarking Erasure codes**
- Designed a **pipeline using Python** for evaluating the **performance** of **Raptor and LT** erasure codes
 - **Simulated** various transmission channels with the **aff3ct** library to accurately emulate **packet loss**
- Spring '24 **Market Index Data Analysis**
- Developed **linear regression** models for 15 market **indicators** to **analyze** underlying hidden **features**
 - Performed **feature engineering** using **trading insights** to **eliminate** several **hyperparameters**
- Spring '24 **Portfolio Optimization**
- Developed a **Python simulator** to efficiently **evaluate** and analyze **Markowitz portfolio optimization**
- Fall '23 **Graph Neural Networks for Optimization**
- Modeled **linear programs** as graphs to solve using **GNNs**, achieving fast and **efficient optimization**
 - Solved a **jigsaw puzzle** using LP and GNN, achieving a **5x speed-up** over **traditional LP solvers**
- Fall '23 **Abstract Protocol Design**
- Built a **verifiable secret sharing** scheme in **self-built** language **mpC**, eliminating formal verification
- Spring '23 **Blockchain simulator**
- Developed a **blockchain** using **SymPy in Python** to analyze **selfish-mining and related attacks**

- Spring '23 **Signal Transformation Tool**
 ○ Designed a **signal processing** tool with **PyQt** and **Tkinter** for **teaching embedded systems**
- Spring '23 **Optimal Data Subset Selection for Model Training**
 ○ **Optimized model training** by **scoring datapoints** to eliminate data **redundancy** over **75%**
- Spring '23 **GCC-like Compiler**
 ○ **Implemented a compiler** for a C-like language **from scratch** using **Flex**, **Bison** and **C++**
- Spring '23 **Postgres++**
 ○ **Enhanced PostgreSQL** by **adding functional dependency** support to **simplify complex schemas**
- Fall '22 **Small-scale Face Detector**
 ○ **Developed a face detector** from scratch using **HoG in Python**, achieving **~2x SOTA** runtimes
- Fall '22 **Custom Animation Rendering Engine**
 ○ **Built a 3D animated world** from scratch in **OpenGL (C++)**, awarded **Best Course Project**
- Spring '22 **Facial Recognition Internship**
 ○ Developed **face recognition** pipelines with **OpenCV**, **Azure**, and **AWS** APIs for **attendance tools**
- Spring '22 **Peer-to-Peer File Sharing App**
 ○ Developed a **reliable P2P network** in **C++** with error handling to ensure **lossless data sharing**
- Spring '22 **RISC in VHDL**
 ○ Implemented a **finite state automaton** in **VHDL**, achieving a functional **CPU computation pipeline**
- Spring '22 **Speech synthesizer**
 ○ **Trained HMMs with Kaldi** for **speech-to-text** transcription, **automating the manual work**
 ○ Developed **text-to-speech synthesizers** with **SoX** and **Python-Coqui** for **Marathi** and **Kannada**
- Fall '21 **Learning Management System Web-App**
 ○ **Built** a remote learning **web app** using **Django**, **SQLite**, and **React** for **fast resource access**
 ○ Created a **Bash CLI tool** to handle **HTTP** and **TCP** requests as a **browser interface alternative**
- Fall '21 **Rapid Visualization of Fractals**
 ○ Developed a **Mandelbrot set visualizer** using **SDL in C++**, enabling **efficient fractal exploration**
 ○ **Optimized** the visualizer tool with **BSTs**, **Heaps**, and **KDTrees** for **~10ms smooth magnification**
- Fall '21 **Generative Pre-training of Pixels**
 ○ Integrated **image generation** into **GPT-2** using **vectorization**, eliminating training of **GANs**
- Summer '21 **Computer Vision Basics**
 ○ Built **CNNs** and **Vanilla NNs** in **PyTorch** for **MNIST digit classification** with **98% accuracy**
- Fall '20 **Visual Maze solver**
 ○ **Solved a visual maze** with **grid extraction** and **graph algorithms** in **~50ms** using **Python**
- Fall '20 **Game Development**
 ○ Developed a **coin-catch game** in **C++** using the **Turtlesim graphics** library at **IIT Bombay**
 ○ Built an **alien shooter game** in **PyGame** collecting **responses** from **~50 users** for **bot development**

Involvement

Mentorships **Tech Mentor** at **XLR8 Robotics competition**, [IIT Bombay](#)
 TAships **Calculus & Embedded Systems**, [IIT Bombay](#), **Algorithms**, [GATech](#)

Skills

Languages & Libraries C/C++, Python, R, JavaScript, SQL, Bash, Huggingface, PyTorch, Django
Software Frameworks Git, MATLAB, Wireshark, PostgreSQL, React.js, Node.js, Android Studio

Extracurriculars

- Member of the **Campus Cats Club** at **GATech**, supporting **initiatives to care for** campus cats
- Amongst **top 10** teams in **Algorithmic Trading** contest organized by **Web and Coding Club**, **IITB**
- Engage in playing the **violin** to enhance musical proficiency, contributing to theatrical **performances**